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Review

Applications of Artificial Intelligence in Dental Malocclusion: A Scoping Review of Recent Advances (2020–2025)

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Abstract

Introduction: Dental malocclusion affects more than half of the global population, causing significant functional and esthetic consequences. The integration of artificial intelligence (AI) into orthodontic care for malocclusion has the potential to enhance diagnostic accuracy, treatment planning, and clinical efficiency. However, existing research remains fragmented, and recent advances have not been comprehensively synthesized. This scoping review aimed to map the current landscape of AI applications in dental malocclusion from 2020 to 2025. Methods: The review followed the Joanna Briggs Institute methodology and the PRISMA-ScR guidelines. The authors conducted a systematic search across four databases (PubMed, Scopus, Web of Science, and IEEE Xplore) to identify original, peer-reviewed research applying AI to malocclusion diagnosis, classification, treatment planning, or monitoring. The review screened, selected, and extracted data using predefined criteria. Results: Ninety-five studies met the inclusion criteria. The majority employed convolutional neural networks and deep learning models, particularly for diagnosis and classification tasks. Accuracy rates frequently exceeded 90%, with robust performance in cephalometric landmark detection, skeletal classification, and 3D segmentation. Most studies focused on Angle's classification, while anterior open bite, crossbite/asymmetry, and soft tissue modeling were comparatively underrepresented. Although model performance was generally high, study limitations included small sample sizes, lack of external validation, and limited demographic diversity. Conclusions: AI offers the potential to support and enhance the diagnosis and management of malocclusion. However, to ensure safe and effective clinical adoption, future research must include reproducible reporting, rigorous external validation across sites/devices, and evaluation in diverse populations and real-world clinical workflows.

Keywords: artificial intelligence; dental malocclusion; deep learning; orthodontics; cephalometric analysis

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1. Introduction

Dental malocclusion, defined as the misalignment between the teeth of the two dental arches as the jaws close, is a clinically significant condition affecting individuals of all ages [1]. With an estimated global prevalence of over 50%, it contributes to esthetic concerns, functional impairments, and psychosocial distress, creating a growing demand

for orthodontic care [2]. Malocclusion is associated with several health complications, including temporomandibular joint (TMJ) disorders, speech difficulties, and an elevated risk for dental caries and periodontal disease [3]. Traditionally, the diagnosis and management of malocclusion relied on clinical examination, cephalometric analysis, dental casts, photographic records, and orthodontic interpretation [4].

In recent years, artificial intelligence (AI) has demonstrated disruptive and transformative potential, particularly in complex pattern recognition and decision-making [5]. Dental and orthodontic providers are increasingly using AI to support and automate diagnostic workflows, enhance treatment planning, improve outcome prediction, and enable more precise monitoring [6]. By applying techniques such as machine learning, deep learning, and computer vision to radiographs, 3D scans, and photographic images, AI models can detect and classify malocclusion with potential gains in accuracy, efficiency, and reproducibility [7].

The application of AI in dentistry encompasses several technological domains. Machine learning enables computers to learn from data and make predictions or classifications without explicit programming [8]. Deep learning, a subset of machine learning, utilizes advanced neural networks such as Convolutional Neural Networks (CNNs), to analyze and interpret complex imaging data [9]. Computer vision seeks to replicate human visual perception in machines, enabling the automated analysis of visual dental inputs, including photographs and cephalometric images [10].

Despite the surge of interest and an increasing volume of published research, the field remains fragmented and poorly synthesized. Studies vary significantly in focus, AI techniques used, and malocclusion subtypes addressed [11]. There is a need for a comprehensive, updated synthesis of evidence to clarify AI's status, identify which tools are truly effective, and outline the challenges that limit clinical translation.

A scoping review is well-suited for systematically mapping a rapidly evolving field across diverse study designs and clinical applications. Current studies span a wide range, from using machine learning to predict craniofacial growth [12], validating AI models for anterior open bite detection [13], and evaluating deep learning-based segmentation tools for orthodontic imaging [14]. However, clinicians often face difficulty when evaluating or adopting these tools due to inconsistent reporting and a lack of comparative validation.

Since 2020, there has been a marked acceleration in AI-driven orthodontic innovation, particularly in the use of deep learning and computer vision for tasks such as cephalometric landmark detection [15] and panoramic image classification for impacted canines [16]. Yet, the field lacks an overview of model reliability, clinical validity, and readiness for real-world integration.

This scoping review aims to fill this critical gap. Its objective is to explore the recent literature (2020–2025) on the use of AI in the diagnosis, classification, treatment planning, and monitoring of dental malocclusion. Specifically, it seeks to map and evaluate the technologies employed, the types and sources of input data, the clinical contexts of AI use, and report performance outcomes. In doing so, this review outlines prevailing trends, highlights areas of intensive development, and identifies research gaps that the field must address to move toward safe and effective clinical translation.

The remainder of this manuscript is organized as follows: Section 2 describes the scoping review methodology, including eligibility criteria, information sources, search strategy, screening, and data extraction. Section 3 presents the study selection results and synthesizes the included evidence across AI methods, clinical tasks, imaging modalities, and reported performance. Section 4 discusses clinical relevance, methodological trends, limitations, and research gaps to support future translation. Additionally, it provides conclusions and recommended directions for clinically reliable and generalizable orthodontic AI.

2. Methods

This scoping review followed the Joanna Briggs Institute methodology for scoping reviews, a structured approach for mapping existing evidence on a given topic [17]. The review also followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) checklist to ensure transparency and methodological clarity [18]. It was registered at the Open Science Framework (osf.io/6jqnd).

2.1. Eligibility Criteria

Table 1 outlines the review criteria. The eligibility criteria included peer-reviewed original research published in English that explored the application of AI in the context of dental malocclusion, including diagnosis, classification, treatment planning, or monitoring. Studies were eligible if they implemented AI technologies (e.g., machine learning, deep learning, neural networks, or computer vision) to analyze or address malocclusion or related orthodontic applications. To be included, studies had to explicitly model malocclusion or an occlusal phenotype/trait (e.g., Angle class, vertical/transverse discrepancies, crowding/spacing, overjet/overbite) as an input, label, or endpoint, or evaluate an orthodontic decision relevant to malocclusion management (e.g., treatment planning/monitoring outcomes). Studies on adjacent conditions (e.g., TMD, airway) were included when malocclusion/occlusion traits were explicitly represented as model inputs and/or outputs, or when the endpoint was explicitly framed within orthodontic malocclusion assessment or treatment-planning pathways. The search excluded review articles, editorials, opinion pieces, case reports, abstracts, dissertations, and grey literature. To ensure the findings reflected current technological capabilities, the authors limited the search to studies published between January 2020 and April 2025.

We restricted inclusion to peer-reviewed original research to prioritize studies with sufficient methodological detail and evaluative reporting for meaningful synthesis. We limited studies to the English language to ensure consistent screening and interpretation of technical methods and outcomes. We selected the publication window (January 2020–April 2025) to reflect current AI capabilities and reporting practices, particularly the rapid expansion of deep learning and computer vision applications in orthodontics. We excluded reviews and other non-original publication types to avoid duplication of evidence and to focus the synthesis on primary model development and evaluation studies. Grey literature was excluded to improve comparability and because reporting of architectures, datasets, and validation procedures is often insufficient for extracting consistent methodological details.

For this review, we defined AI as computational methods that (i) learn patterns from data (machine learning/deep learning) or (ii) provide algorithmic decision support through explicit knowledge/rule representations (expert systems). Studies that were purely deterministic measurement tools without inference or decision logic were excluded. When commercial platforms did not disclose model architecture, we included them only if the study described an AI inference capability (e.g., automated detection/classification/segmentation or prediction) and reported a validation comparison.

Table 1. Inclusion/exclusion criteria.

Inclusion Criteria	Exclusion Criteria
• Human subjects' dental applications • Peer-reviewed original articles written in English • Articles published between January 2020 and April 2025 • Focused on AI and Dental Malocclusion	• Conference proceedings • Opinion articles (letter to the editor, editorial comments, opinion) • Literature reviews (narrative, scoping, systematic, meta-analysis) • Animal studies • Abstracts without full text • Case Reports

2.2. Information Sources and Search Strategy

The review searched the literature across four major databases: PubMed, Scopus, Web of Science, and IEEE Xplore (Table 2). The authors selected these databases to capture both the medical/dental literature and technical/engineering literature. Four authors (MH, OC, NB, CZ) collaborated with a research librarian (CS) using a combination of controlled vocabulary (e.g., MeSH terms in PubMed) and free-text keywords. The search terms targeted three main concepts: (1) AI (e.g., "artificial intelligence", "machine learning", "deep learning", "neural networks", "computer vision"), (2) malocclusion and related terms (e.g., "malocclusion", "dental malocclusion"), and (3) imaging or data modalities (e.g., "radiograph", "cephalometric", "photograph", "intraoral scan"). Boolean operators ("AND", "OR") were used to combine these terms appropriately. Table 2 outlines the search strategy.

To adapt the strategy across databases, we translated controlled vocabulary (e.g., MeSH in PubMed) into database-specific subject headings where applicable and adjusted field tags (e.g., TITLE-ABS, TS) and wildcarding. We iteratively tested synonym blocks for AI terms, malocclusion/occlusion terms, and dental/orthodontic context terms to ensure consistent recall across platforms (Table 2). Backward and forward citation tracking of included studies was also performed to reduce the risk of missed eligible studies.

Table 2. Search strategies.

Database & Search Date	Search Strategy & Filters	Number of Studies Found
PubMed 14 April 2025	("Artificial Intelligence" [Mesh] OR "Artificial Intelligence" [tiab] OR "AI" [tiab] OR "Machine Intelligence" [tiab] OR "Machine Learning" [tiab] OR "Computer Vision" [tiab] OR "Deep learning" [tiab]) AND ("Malocclusion" [Mesh] OR "Malocclusion" [tiab] OR "Occlusion" [tiab]) AND ("Dent*") Filters: English & Publication Date	124
Scopus 14 April 2025	TITLE-ABS (("Artificial Intelligence" OR "AI" OR "Machine Intelligence" OR "Machine Learning" OR "Computer Vision" OR "Deep learning") AND ("Malocclusion" OR "Occlusion")) AND ALL (("Dent*")) AND PUBYEAR > 2019 AND PUBYEAR < 2026 AND (LIMIT-TO (LANGUAGE, "English")) Filters: English & Publication Date	197
Web of Science 14 April 2025	(TS = (("Artificial Intelligence" OR "AI" OR "Machine Intelligence" OR "Machine Learning" OR "Computer Vision" OR "Deep learning") AND ("Malocclusion" OR "Occlusion"))) AND ALL = (("Dent*")) Filters: English & Publication Date	112
IEEE Xplore 14 April 2025	("Artificial Intelligence" OR "AI" OR "Machine Intelligence" OR "Machine Learning" OR "Computer Vision" OR "Deep learning") AND ("Malocclusion" OR "Occlusion") AND ("Dent*") Filters: English & Publication Date	25

2.3. Study Selection

All search results were imported into the EndNote reference management software Version 21, and duplicates were removed. Three reviewers (OC, NB, CZ) independently screened the titles and abstracts using the predetermined inclusion criteria. The same reviewers retrieved the full-text versions of the screened studies and independently assessed the articles for inclusion. The reviewers resolved inclusion disagreements through discussion, and if they could not achieve a consensus, a fourth reviewer (MH) made the

final decision. Figure 1 summarizes the study selection process, including reasons for exclusion at the full-text screening stage.

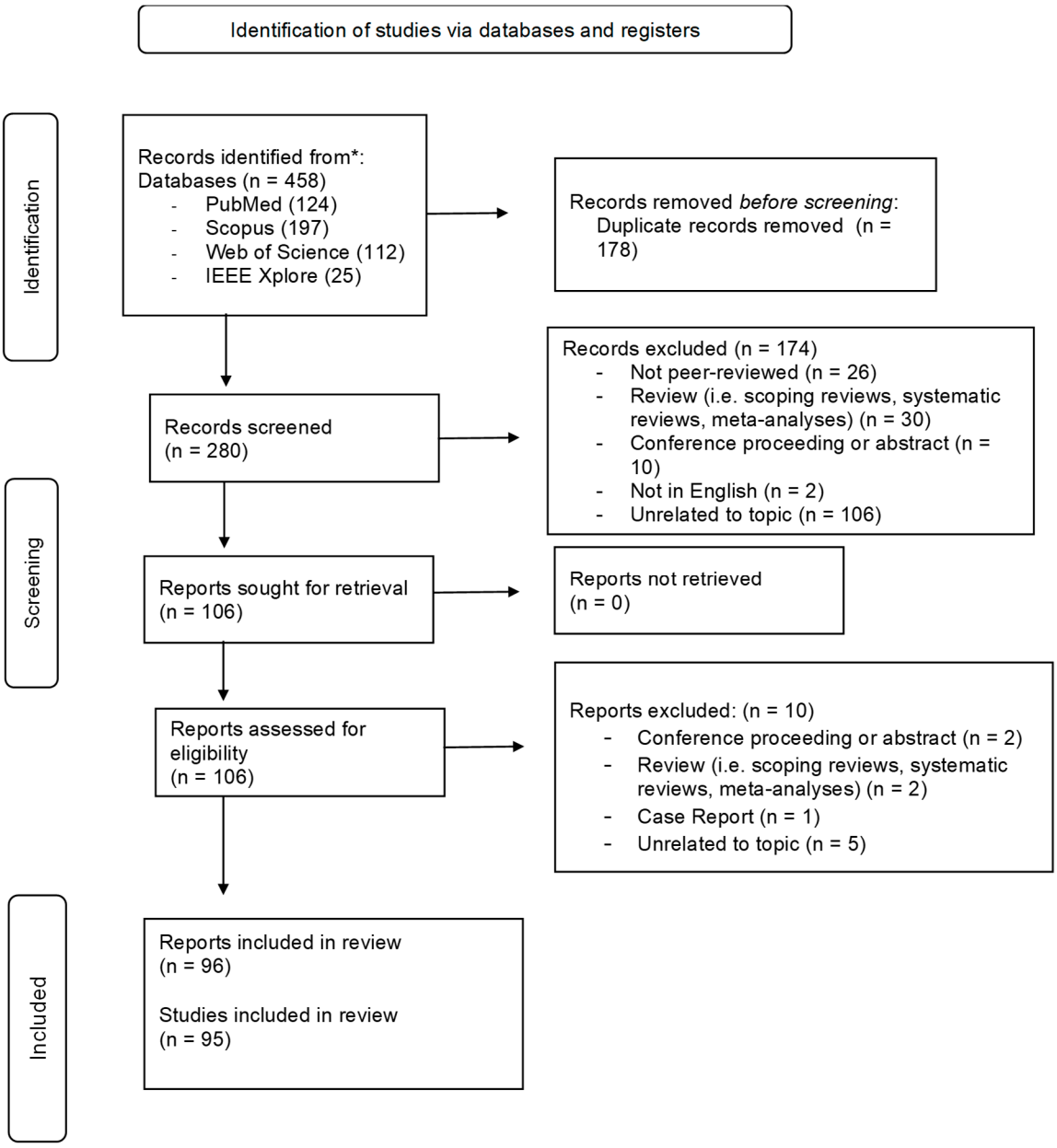

Figure 1. PRISMA flow chart of article selection.

2.4. Data Extraction

Three reviewers (OC, NB, CZ) independently extracted data from the included studies using a structured form developed in Microsoft Excel. The form included the following data items: authors, year of publication, country of origin, study design, data sources (e.g., radiographs, photographs, 3D scans), type of malocclusion addressed, specific AI methodology used, purpose of AI application (e.g., diagnosis, treatment planning), training and validation approach, and outcome measures (e.g., accuracy, sensitivity, specificity, F1 score, and study population). Additionally, the reviewers noted any reported limitations, ethical considerations, or practical challenges related to the AI application. To ensure accuracy and consistency, a fourth reviewer (MH) resolved discrepancies and conducted random quality checks on the extracted data.

2.5. Data Synthesis

The authors analyzed the extracted data using both quantitative and qualitative techniques. A descriptive numerical summary characterized the study distribution by year, country, AI type, and clinical application. Tabular presentations highlighted the types of AI models and malocclusion conditions studied. For the qualitative component, the authors conducted a thematic narrative synthesis, systematically reviewing the qualitative findings, identifying recurring concepts, grouping these concepts into broader themes, and then interpreting these themes in the context of the research aims. This approach identified emerging patterns, common AI applications, performance metrics, methodological trends, and key research gaps and underexplored areas. Given heterogeneity in reported metrics, datasets, and validation designs, performance findings were synthesized descriptively rather than pooled quantitatively.

3. Results

3.1. Overview of Study Selection

The search initially yielded 458 records across four databases: PubMed (124), Scopus (197), Web of Science (112), and IEEE Xplore (25). After removing 178 duplicate entries, 280 unique records remained for title and abstract screening. Abstract review excluded 174 additional studies. Reasons for exclusion included: 106 studies were unrelated to malocclusion, 30 were review articles or meta-analyses, 26 were not peer-reviewed original research, 10 were conference abstracts, and 2 were non-English, leaving 106 full-text reports. The full-text article review excluded an additional 10 articles; five were off-topic, two were conference abstracts, two were reviews, and one was a case report. Ultimately, the review identified 95 eligible research studies (Appendix A). The PRISMA-ScR flow diagram in Figure 1 outlines the full study selection process.

3.2. Publication Trends and Geographic Distribution

The number of publications increased over the review period, reflecting growing interest in the application of AI to dental malocclusion (Table 3). The search identified 16 studies published between 2020 and 2021 and 26 studies from 2022 to 2023. The most significant surge occurred between 2024 and early 2025, with 53 studies published. This upward trend highlights the expanding research activity in this field. Geographically, China contributed the most with 28 studies, followed by South Korea (9), the USA (8), Germany (5), Saudi Arabia (5), Australia (4), Turkey (4), India (3), and Iraq (3) (Table 3).

Table 3. Distribution of studies by year and country.

Years	Total Number of Studies	Top Contributing Countries: Number of Studies
2020–2021	16	South Korea: 5, Saudi Arabia: 1, Australia: 1, Taiwan: 1, Turkey: 1, USA: 1, Germany: 2, India: 1, Egypt: 1, UAE: 1, Slovakia: 1
2022–2023	26	China: 8, Saudi Arabia: 1, Germany: 2, South Korea: 3, Turkey: 1, Malaysia: 1, USA: 5, Iraq: 2, Denmark: 1, Australia: 1, Iran: 1
2024–2025	53	Egypt: 1, Saudi Arabia: 3, Iraq: 1, Brazil: 2, Iran: 1, Turkey: 2, China: 20, India: 2, Taiwan: 3, Australia: 2, Colombia: 1, Japan: 2, Sweden: 1, South Korea: 1, Israel: 1, Denmark: 1, USA: 2, UAE: 1, Thailand: 2, Italy: 2, Germany: 1, UK: 1

3.3. AI Techniques Used in Malocclusion Research

The studies encompassed a wide range of AI methodologies, with convolutional neural networks (CNNs) being the most frequently used technique, featured in 36 studies

(37.9%) (Table 4). Thirty-five studies (36.8%) used traditional machine learning approaches, including support vector machines (SVM), random forest (RF), and k-Nearest neighbors. Other studies utilized deep learning models that did not fall strictly under CNN architectures (15 studies, 15.8%). Only 1.1% of studies employed hybrid or ensemble models, while 8 studies (8.4%) used alternative or unspecified AI techniques (Table 4).

Table 4. Artificial intelligence types and frequency.

AI Type	Number of Studies	% of Total
CNN	36	37.9%
Support Vector Machine, Random Forest	35	36.8%
Deep learning	15	15.8%
Hybrid/Ensemble methods	1	1.1%
Others	8	8.4%
Total	95	100.0%

3.4. Clinical Applications of AI

The diagnosis and classification of malocclusion was the most common clinical AI application, appearing in 55 studies (Table 5). These studies focused on determining skeletal classes, detecting cephalometric landmarks, and identifying features such as overjet and overbite. Twenty-seven studies used AI for treatment planning, where it assisted in generating orthodontic or surgical recommendations based on diagnostic inputs. Twenty-six studies assessed severity, typically quantifying the degree of malocclusion or dental crowding. A smaller group of three studies applied AI to predict longitudinal craniofacial growth patterns, helping inform early intervention strategies. Several studies reported on specific tools or model families used for these applications, including WebCeph, 3D Slicer, and EfficientNetV2, reflecting the integration of proprietary and open-source platforms in AI-based orthodontic research. Because some studies addressed more than one clinical task, counts across application categories are not mutually exclusive.

Table 5. Clinical application of artificial intelligence in malocclusion studies.

Clinical Application	Number of Studies *	Common AI Models Used
Diagnosis/classification	55	Deep learning-based segmentation models Convolutional Neural Networks (CNNs) WebCeph software
Treatment planning	27	WebCeph software 3D Slicer software Deep learning with transformer architectures
Severity assessment	26	EfficientNetV2 and DenseNet201 Custom CNNs YOLO-based models
Growth prediction	3	Vision Transformer Self-supervised pre-training architectures Hybrid deep learning models

* Some studies contributed to more than one clinical application category (i.e., categories are not mutually exclusive), and some studies used more than one model.

3.5. Types of Malocclusion Conditions Studied

Angle's classification system (Class I, II, and III) was the most frequently used malocclusion framework, referenced in 56 studies (Table 6). Other malocclusion types received more limited focus: just two studies [19,20] addressed anterior open bite, while 10 studies [21–30] explored crossbite, crowding, and asymmetry. Thirty studies either did not specify a malocclusion category or referred to mixed/generalized cases, indicating variability in diagnostic focus and inconsistent reporting across the literature. Because some studies addressed more than one condition, counts are not mutually exclusive.

Table 6. Malocclusion condition studied.

Condition Category	Number of Studies *	Notes
Angle classification (I/II/III)	56	Includes any mention of Class I, II, or III malocclusion.
Anterior open bite	2	Specifically mentions of open bite.
Crossbite, crowding, asymmetry	10	Includes references to spacing, crowding, asymmetry, or crossbite.
Mixed/Other	30	Studies with general or unspecified malocclusion descriptions or where classification was unclear.

* Some studies addressed more than one condition.

3.6. Key Themes and Patterns Identified

A thematic synthesis revealed that the studies employed AI technologies to detect malocclusion types, classify treatment needs, analyze cephalometric landmarks, predict soft and hard tissue changes, segment dental structures, and support surgical planning (Table 7). Predictive targets frequently included overjet, incisor position, dental crowding, and Class III skeletal discrepancies. Three studies [31–33] used AI for early screening from profile photographs or intraoral images, particularly for Class III or deepbite conditions. Notably, a small subset of models [14,31,34] used multiple input modalities, such as CBCT and facial photographs, to enhance diagnostic performance. Overall, most studies focused on image-based diagnostic workflows, whereas fewer studies evaluated downstream outcomes (e.g., treatment planning impact or longitudinal prediction).

Table 7. Summary of themes, performance, and gaps.

Category	Key Findings
Emerging patterns	• Artificial intelligence (AI) models were used to detect malocclusion types, classify treatment needs, analyze cephalometric landmarks, predict soft and hard tissue changes, segment dental structures, assess facial symmetry, and assist in surgical planning [35]. • Overjet, incisor changes, dental crowding, and skeletal discrepancies (especially Class III) appeared as targets of prediction or analysis [32]. • AI supported early detection through profile images or photographs, and several studies focused specifically on Class III and deepbite malocclusions [36]. • Some models integrated multiple modalities [14] (e.g., CBCT + profile photos), while others used only facial images or intraoral scans.

Table 7. *Cont.*

Category	Key Findings
Performance metrics	Accuracy Most AI tools achieved 80–95% accuracy [37,38], with multiple models exceeding 90% for malocclusion classification, cephalometric tracing, and 3D segmentation [39,40]. Top accuracy results included: • ResNet18: 96.5% (crossbite detection) [27] • Inception V3: 92.6% [16] • MobileNetV3 (with enhancement): 90.7% [26] • 3D segmentation (Dice score): 95.1% [41] • Malocclusion classification: ~93.1% (molars) [42] • WebCeph agreement with clinicians: ICC = 0.90 Models analyzing hard tissue landmarks or simpler classification problems [15] were more accurate than those working with soft tissue or mixed cases [43]. Sensitivity & Specificity Sensitivity and specificity were generally high, especially for clear-cut skeletal Class III and crossbite cases. Examples include: • ResNet18: Sensitivity 89%, Specificity 99.2% [27] • Inception V3: Sensitivity 93.6%, Specificity 91.3% [16] • TMD model: Sensitivity 71.2%, Specificity 84.4% [44] • Class III screening from profile photos: Sensitivity 60%, Specificity 92.5% [45] Sensitivity was lower for soft tissue tasks or borderline cases [37], while specificity remained strong overall [46]. Precision Several models showed high precision, especially those with preprocessing or task-specific architectures [38]. Reported values include: • ResNet18: 97.8% [38] • Inception V3: 93.6% [16] • MobileNetV3 (with CLAHE): top performer in precision among its group [26] • Malocclusion (molars and canines): ~88.7% [42] • TMD model (external test): F1 = 0.75 [44] Precision tended to be lower in ambiguous malocclusion types (e.g., Class I) [44] and higher for clearly defined skeletal cases [44]. Stability Many models demonstrated consistent performance across validation folds and test sets [47]. • Studies using 5-fold or 10-fold cross-validation showed minimal performance variation [41]. • WebCeph and DSS tools had stable agreement rates across subgroups [19]. • Growth and symmetry models maintained accuracy across bootstrap or external validations [48,49]. Models using image enhancement (e.g., CLAHE) and hybrid training strategies tended to report better stability [26].
Overall Outcomes	Across studies, AI tools consistently supported accurate diagnosis, faster workflows, and reduced manual effort [12,50–53]. • Many models outperformed or matched junior clinicians [24,54] • Tools proved especially valuable in screening, cephalometric analysis, surgical planning, and underserved settings [33] • Tasks involving 3D segmentation, skeletal Class III detection, and landmark tracing were the most successful [47,54–63] Performance declined in soft tissue, borderline cases, or when there was limited training data [64].

Table 7. *Cont.*

Category	Key Findings
Methodological trends	• Deep learning models such as ResNet18, MobileNetV3, Inception V3, and DenseNet were commonly used [26]. • Many studies enhanced performance through image preprocessing (e.g., CLAHE), augmentation, and hybrid model design [25]. • Interpretability was addressed by using GradCAM, SHAP, and VisualBackProp [47,65,66]. • Data sources included cephalograms, CBCT, intraoral images, and profile photos [30,41,67–69]. • Decision support tools (e.g., DSS, nomograms) were developed to translate AI predictions into actionable clinical guidance [19].
Evidence Gaps	• Performance declined in borderline malocclusions (e.g., Class I), soft tissue predictions, and extraction decision modeling [60]. • Studies often had small samples (e.g., 34–66 patients), limited to single centers, and lacked external validation [70–74]. • Most models developed for specific ethnic groups (e.g., Chinese, Korean, Saudi), limiting their generalizability [48,75]. • Manual landmarking required in many applications [76,77]. • Soft tissue accuracy lagged hard tissue modeling [50]. • Long-term outcome predictions rarely addressed.
Recommendations	• Improvements should focus on better handling of borderline and soft tissue cases. • Integration into clinical workflows is a priority. • Improved interpretability tools will help gain clinician trust and enhance usability in everyday practice.

3.7. Performance of AI Models

Most AI models demonstrated robust performance across diagnostic and analytical tasks (Table 7). Reported accuracy typically ranged from 80% to 95%, with several models exceeding 90% in malocclusion classification, cephalometric landmark detection, and 3D segmentation [16,25,27,37,78,79]. For example, ResNet18 [27] achieved 96.5% accuracy in detecting crossbite, while Inception V3 [16] reached 92.6%. WebCeph [79] demonstrated strong agreement with clinicians, with an intraclass correlation coefficient (ICC) of 0.90.

Sensitivity and specificity scored high, especially for clearly defined conditions such as skeletal Class III and crossbite. One model [27] using ResNet18 reported 89% sensitivity and 99.2% specificity. In contrast, tasks involving soft tissue analysis or borderline cases had lower sensitivity, though specificity remained consistently high. Precision was also high for most models, with values such as 97.8% for ResNet18 and 93.6% for Inception V3 [16]. Although the studies did not consistently report F1 scores, the models that incorporated task-specific architectures and advanced preprocessing methods tended to report the highest-performing results.

To improve interpretability, we summarized performance descriptively and stratified reporting by imaging modality and by malocclusion category/clinical task, where sufficient information was provided. By imaging modality, performance reporting was most frequently derived from cephalograms and CBCT, with additional evidence from facial photographs and intraoral images; however, differences in acquisition protocols, devices, preprocessing pipelines, and reported metrics limited direct comparability across modalities. By malocclusion category/clinical task, performance evidence was strongest for common classification problems (e.g., Angle Class I–III and skeletal pattern classification), cephalometric landmark detection, and segmentation, whereas fewer studies reported performance for less common phenotypes and more complex tasks (e.g., soft tissue analysis and borderline cases). Accordingly, reported performance should be interpreted in the context of dataset composition, validation design, and the specific modality/task evaluated.

Cross-validation methods confirmed model stability. Studies using five- or ten-fold cross-validation showed low variance across folds. WebCeph demonstrated stable agreement rates across different demographic groups, and image enhancement techniques like CLAHE were linked to improved consistency and robustness. Nevertheless, heterogeneity in validation approaches and incomplete reporting of subgroup performance constrained quantitative synthesis and limited the ability to compare robustness across modalities and malocclusion subtypes.

3.8. Methodological Trends

Across the included studies, deep learning frameworks such as ResNet18, DenseNet201, Inception V3, and MobileNetV3 were the most widely used architectures (Table 7). Performance improvements were commonly attributed not only to architecture choice but also to optimization decisions, including preprocessing, augmentation, and hyperparameter selection. Preprocessing techniques (e.g., CLAHE), data augmentation, and hybrid learning strategies were frequently used to improve model performance. However, reporting of these optimization steps was heterogeneous, and many studies did not clearly describe whether tuning was prespecified, how hyperparameters were selected, or how experimental safeguards against data leakage were implemented (e.g., ensuring patient-level separation between training/validation/test sets and applying preprocessing within the training fold only). In robust experimental design, optimization procedures should be reproducible and systematically evaluated using a defined protocol (e.g., fixed data splits or nested cross-validation) rather than ad hoc trial-and-error. Structured optimization frameworks drawn from design-of-experiments, including parametric orthogonal-array approaches such as the Taguchi method [80], can reduce the search burden while enabling systematic, transparent tuning of preprocessing parameters and model hyperparameters.

The AI tools GradCAM, SHAP, and VisualBackProp addressed interpretability. Data sources included cephalograms, CBCT scans, intraoral images, and profile photographs, with several studies developing decision-support systems to provide clinically relevant outputs. Overall, clearer documentation of optimization workflows (preprocessing pipelines, hyperparameter search strategy, stopping criteria, and leakage controls) would improve reproducibility and strengthen confidence in reported performance gains.

3.9. Evidence Gaps and Limitations

Despite promising results, the review identified several gaps in evidence and limitations. AI models performed less accurately in borderline clinical cases, particularly Class I malocclusion, where diagnostic ambiguity reduced precision (Table 7). Performance also declined in soft tissue-related tasks, which remain challenging due to anatomical variability and lower contrast in imaging. Few studies addressed long-term treatment outcome prediction, limiting insights into the durability of AI-assisted planning. Methodologically, many studies relied on small and demographically homogeneous datasets, often from a single center, reducing generalizability. A widespread lack of external validation further restricted confidence in model robustness. Additionally, several models needed manual annotations of cephalometric landmarks, limiting scalability and increasing labor intensity.

4. Discussion

4.1. Summary of Findings

This review highlights a rapid acceleration in orthodontic AI research since 2020, reflecting growing enthusiasm for integrating automated tools into clinical practice [81]. CNNs and deep learning approaches dominated the methodological landscape, owing to their strong image-processing capabilities and adaptability. Most studies examined diag-

nostic and classification tasks involving Angle's malocclusion classification (Class I, II, III), where AI models demonstrated high clinical relevance and technical accuracy. Reported accuracy rates frequently surpassed 90%, with tools such as CephNet, Cephio, and WebCeph achieving particularly strong performance in specialized tasks such as cephalometric landmark detection and 3D segmentation. These findings underscore the clinical potential of AI in improving diagnostic consistency, reducing manual effort, and expanding access to orthodontic care. However, because performance metrics and validation strategies were inconsistently reported across studies, these high accuracies should be interpreted in light of study design, dataset size, and the presence (or absence) of external validation.

4.2. Comparison with Prior Reviews or Traditional Methods

Unlike previous reviews that broadly surveyed AI use in dentistry or included mixed study types, this scoping review synthesized research specifically designed for malocclusion applications [82–84]. By narrowing the search to studies published between 2020 and 2025, this review captured recent advancements in deep learning, machine learning, and computer vision applied to orthodontics. The exclusion of reviews, abstracts, and grey literature enhanced methodological consistency and clinical relevance. This narrower scope enabled a more precise mapping of AI trends, tools, and knowledge gaps, offering new insights that extend earlier, more generalized analyses.

Compared to traditional manual approaches [85], such as cephalometric tracing and skeletal classification by orthodontists, AI methods offer substantial advantages in speed, scalability, and reproducibility [86–90]. Deep learning models, particularly CNNs, can process radiographs and 3D scans rapidly and provide consistent outputs without fatigue or inter-operator variability [19,56,91,92]. Accuracy matched or exceeded junior clinicians, particularly in landmark detection and skeletal Class III classification [54]. Nonetheless, clinical translation depends not only on accuracy but also on reproducibility, generalizability across sites/devices, and integration into real-world workflows.

However, significant challenges remain. Model interpretability is often lacking, making it difficult for clinicians to trust or understand AI-generated results. Generalizability is another key concern, as models may perform well in internal validation but struggle when applied to external datasets, different imaging protocols, or different devices. Taken together, these limitations highlight the need for transparent reporting, rigorous external validation, and clinically grounded implementation pathways before widespread adoption.

4.3. Methodological Considerations in the Reviewed Studies

A methodological strength across the reviewed studies was the consistent use of advanced deep learning models, including architectures like ResNet, DenseNet, and MobileNet, which enabled high performance in image analysis tasks. Several studies also explored more clinically realistic workflows by integrating multiple imaging modalities (e.g., combining cephalograms, CBCT, and facial photographs), which may better reflect clinical decision-making and case complexity [78].

However, notable limitations were observed. Many studies relied on small, single-center datasets with limited demographic diversity, reducing the generalizability of their findings. Most studies failed to perform external validation, raising questions about a model's robustness in real-world practice. Furthermore, workflows often required manual preprocessing steps, such as image cropping, landmark annotation, or quality filtering, limiting their full automation and scalability in clinical settings.

Methodological heterogeneity remains a key limitation. Studies varied widely in reporting of architectures, dataset composition, train/validation/test splitting strategies, and metrics, making direct comparisons difficult and limiting reproducibility. To address

this, we propose a minimal reporting checklist tailored to orthodontic AI research (Table 8), adapted from established AI reporting frameworks (CONSORT-AI [93], TRIPOD-AI [94]) and imaging-focused guidance (CLAIM [95]), emphasizing the minimum information needed to interpret validity, bias, and clinical readiness. Although Table 8 is presented as an orthodontic-AI checklist, its domains (e.g., reference standard, data provenance, external validation, subgroup performance, and workflow evaluation) reflect broadly applicable principles for trustworthy clinical AI. As such, the checklist can be readily adapted to other areas of dentistry and broader healthcare, as well as to related applied AI settings where transparency, generalizability, and safety-oriented evaluation are required.

Table 8. Minimum reporting checklist for orthodontic AI and related clinical AI studies.

Checklist Domain	Minimum Item to Report	Examples/Notes
Clinical aim	Intended use and clinical task	Screening vs. diagnosis vs. planning vs. monitoring; target malocclusion traits.
Population	Eligibility criteria and phenotype definitions	Define Angle class and other traits (overjet/overbite, crowding, asymmetry).
Data provenance	Source, centers, dates, and sample size	Single vs. multi-center; number of patients/images; missing data handling.
Demographics	Age/sex/ethnicity distribution	Report and justify; note representativeness of target population.
Imaging protocol	Device/protocol metadata and acquisition settings	Modality, vendor/model, geometry, resolution; exposure where applicable.
Reference standard	Labeling process and annotator expertise	Who labeled, guidelines used; inter-/intra-rater reliability.
Preprocessing	Preprocessing and augmentation pipeline	Normalization/CLAHE; augmentation types; applied only on training folds.
Data splitting	Train/validation/test strategy and leakage controls	Patient-level split; nested cross-validation when tuning; no overlap across splits.
Model details	Architecture and training configuration	Backbone, loss, optimizer, epochs, batch size; initialization/transfer learning.
Hyperparameters	Tuning strategy and search space	Grid/random/Bayesian; stopping rules; design of experiment/Taguchi as systematic option.
Evaluation	Primary metrics with uncertainty	Task-appropriate metrics, confidence intervals; calibration for probabilistic outputs.
External validity	External and cross-domain validation	Testing across sites/devices; domain shift analysis; harmonization/adaptation methods.
Subgroup analysis	Phenotype- and demographic-stratified performance	Rare phenotypes; fairness across demographics; error analysis and failure modes.
Interpretability & safety	Explanation method and clinical use	Grad-CAM/SHAP; how explanations are presented; safety monitoring plan.
Clinical workflow	Prospective workflow/user evaluation	Usability, chair time, acceptance; impact on decisions; reporting of adverse events.

Terminology was also used inconsistently across the literature. To support clearer future reporting, we recommend a standardized taxonomy distinguishing: (i) machine learning (non-neural models), (ii) deep learning (neural models), (iii) CNNs (deep learning subtype for images), (iv) transformers/ViTs (deep learning subtype), (v) hybrid models (e.g., two-stage, multimodal fusion, or physics-informed approaches), and (vi) decision support systems (learning-based vs. rule-based). Standardizing definitions would reduce

ambiguity in synthesis and improve cross-study interpretability. We also recommend that future studies explicitly label the model family and pipeline type in the Methods (e.g., ML vs. DL; CNN vs. transformer; unimodal vs. multimodal; end-to-end vs. two-stage) to support clearer synthesis and reduce inconsistent terminology across publications.

4.4. Clinical Implications

AI is poised to reshape orthodontic workflows by enabling faster, more consistent, and more objective diagnostic processes. Models trained on cephalograms, CBCT scans, and clinical photographs can produce rapid assessments, reduce clinician workload, and minimize diagnostic variability. Tools like WebCeph are already in use, offering real-time support for automated tracing, classification, and treatment simulation.

Importantly, AI applications show promise for remote screening, particularly beneficial in underserved or rural areas with limited access to orthodontic specialists. However, clinical reliability may be reduced for underrepresented malocclusion phenotypes. Only a small number of studies addressed anterior open bite, crossbite, or facial asymmetry, which increases the risk of spectrum bias and limits confidence when these models are applied to less common presentations. Because most studies did not report phenotype-stratified performance (e.g., rare phenotype sensitivity/specificity, confidence intervals, and error modes), this review could not quantify the impact of sparse phenotype coverage on model bias; nevertheless, the clinical implication is clear: Models trained predominantly on common phenotypes may underperform for rare or complex cases, potentially increasing misclassification risk and inappropriate treatment planning.

Interpretability is also a practical clinical need rather than a purely technical preference. Explanations can support error detection in borderline cases, calibrate clinician trust, document rationale for accountability, and facilitate shared decision-making with patients. Interpretability methods do not require a "special needs interpreter"; rather, they require clinically designed interfaces and appropriate user training so explanations are understandable, standardized, and not misleading.

Finally, although several studies proposed decision support systems, few evaluated real-world outcomes such as workflow impact, chair time, usability, or user acceptance. Future work should include prospective workflow studies assessing operational benefit and clinical safety, including clinician interaction, failure handling, and user experience.

4.5. Limitations

This scoping review followed established methodological guidance, which does not require a formal quality appraisal of included studies. However, while this approach prioritizes breadth over depth and effectively captures the scope, diversity, and evolving nature of AI applications in malocclusion research, the absence of a structured risk-of-bias assessment warrants caution when interpreting the strength of individual study findings and the certainty of reported performance claims.

Limiting the review to English-language publications between January 2020 and April 2025 may miss relevant studies in other languages or published before 2020. Nonetheless, the review deliberately chose this period to focus on the most current and technologically advanced developments, particularly the rapid rise in deep learning and computer vision tools in clinical orthodontics.

Finally, while most studies offer insight into AI performance, some lacked comprehensive reporting on technical aspects such as model architecture, training parameters, or validation design. This limited the ability to perform direct methodological comparisons. Most importantly, limited reporting of subgroup performance (by phenotype and demographics) and limited external validation constrained our ability to quantify bias,

fairness, and real-world generalizability across clinical settings and patient populations. Despite these limitations, the review provides a timely and meaningful synthesis that lays the groundwork for future focused evaluations, interdisciplinary research, and clinical translation.

4.6. Research Gaps and Future Directions

Future research should address several notable gaps identified in the current literature. Certain malocclusion types, such as anterior open bite, crossbite, and facial asymmetry, remain underrepresented. Addressing these gaps will require deliberate sampling strategies and routine reporting of phenotype-specific errors, particularly in borderline cases and less common subtypes.

AI performance also remains suboptimally studied in borderline cases, particularly Class I malocclusion, and in soft tissue modeling, where anatomical variability and imaging limitations introduce additional challenges. Soft-tissue-related tasks frequently appear less robust than hard-tissue tasks, and demographic homogeneity (ethnicity, age, sex) combined with geographic concentration of datasets likely worsens generalizability. Because demographic stratification and subgroup performance were often not reported, we could not quantify the effect; however, future studies should routinely report subgroup performance and calibration, especially for soft-tissue endpoints.

Very few studies explored long-term treatment outcomes, an area crucial for advancing personalized orthodontic care. Future "long-term prediction" studies should prioritize clinically meaningful outcomes such as relapse/stability (alignment and occlusal indices), retention outcomes, long-term skeletal and soft tissue changes, stability of planned tooth movement, and patient-centered outcomes where available.

A major gap across the literature is the lack of external validation. Practical strategies to improve generalizability include multi-center external testing, cross-device and cross-protocol evaluation, harmonization of imaging pipelines, domain adaptation approaches, federated learning across institutions, preregistration of evaluation protocols, and curated reporting of device/protocol metadata to enable domain-shift analysis. In addition, hybrid simulation–AI approaches [96] (e.g., combining physics-based modeling with AI) can support controlled generation of synthetic data to augment underrepresented conditions, reduce overfitting, and strengthen external validity testing (consistent with emerging examples in the broader biomedical AI literature). In imaging-based orthodontic AI, cross-device and cross-protocol testing is a clinically essential form of external validation because equipment and acquisition differences can produce meaningful domain shift.

Most studies still rely on unimodal inputs (single imaging modality). Future research should more explicitly pursue multimodal fusion approaches (e.g., combining cephalograms, CBCT, facial photographs, and intraoral scans) to better reflect clinical decision-making and improve model robustness and clinical usefulness.

Finally, future work should focus on developing interpretable AI systems that integrate seamlessly into clinical workflows, enabling clinicians to act on AI-generated recommendations with greater confidence. Clinical translation will require not only technical validation but also prospective evaluation in real-world settings, including usability, chair time reduction, clinician acceptance, and safety-oriented reporting consistent with early-stage clinical evaluation guidance for decision support tools. These studies are essential to demonstrate real-world value beyond retrospective performance metrics and to support responsible clinical adoption.

4.7. Conclusions

AI holds significant promise for enhancing the diagnosis, classification, and treatment planning of dental malocclusion. With high accuracy and the potential to streamline clinical workflows, AI tools can augment orthodontic care by increasing diagnostic consistency, reducing manual workload, and expanding access to underserved populations. However, to realize this potential, future efforts must prioritize robust external validation, model interpretability, and seamless integration into clinical settings. To support clinical translation, future work must move beyond high internal accuracy toward reproducible reporting, rigorous external validation across sites and devices, explicit evaluation of rare phenotypes and demographic subgroups, and prospective workflow trials demonstrating real-world clinical value. In addition, future studies should adopt standardized terminology and minimum reporting practices to enable meaningful comparison across datasets, devices, and clinical contexts. Ultimately, demonstrating clinical readiness will require prospective evaluation in representative populations, transparent documentation of performance limitations, and evidence that AI tools improve decision-making and efficiency without compromising patient safety. Advancing the field will require close collaboration between clinicians, data scientists, and engineers to ensure that AI systems are both technically sound and useful in real-world orthodontic practice.

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Appendix A. List of the 95 Reviewed Studies

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[12]	16–45 years old	Machine learning (non-neural)-Random forest regression	Class III malocclusion	Overjet was the key predictor of craniofacial changes; most cephalometric measures showed significant postsurgical changes except SNA, A to NP, overbite, and lower lip to E-plane.	Overjet was a major predictor of post-surgical changes, strongly forecasting shifts in overjet and overbite.
[13]	Age not reported	Clinical software (AI-assisted)	Anterior Open Bite	The software efficiently diagnosed anterior open bite with 82.2% agreement to conventional methods, offering greater comprehensiveness and paving the way for future AI advancements.	The DFO software showed 82.2% agreement with conventional diagnosis (CFO); considered more comprehensive and reliable than CFO.
[14]	15–30 years old	Deep learning-based segmentation models	Class I malocclusion	AI-based segmentation using CephX provides valid and reliable 3D tooth models. The authors recommended its clinical use in mild malocclusion cases where restorations are minimal	The AI-generated 3D tooth models showed high agreement with IOS and CBCT images. AI-based segmentation using CephX was efficient, reliable, and clinically applicable
[50]	12–14 years old	Clinical software (AI-assisted)	Class III malocclusion	AI-based segmentation with CephX yields valid, reliable 3D tooth models and is recommended for clinical use in mild malocclusions with minimal restorations.	AI-driven analysis using Webceph was effective in detecting morphological differences
[15]	13–14 years old	Commercial AI (undisclosed)	Classes I–III malocclusion	Right mandibular body length is a key marker for Class III malocclusion, with right-sided mandibular deviation more common. Webceph AI showed excellent repeatability (ICC > 0.97) and can assist in classification and aesthetic evaluation by detecting subtle asymmetries.	The Webceph AI tool reliably identified right mandibular body length and deviation direction as key malocclusion markers in Saudi females.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[16]	Age not reported	Machine-learning-based AI model	Mild, moderate, and severe malocclusion cases	AI models moderately predict orthodontic treatment outcomes, performing better for mild-to-moderate cases, and should complement clinical expertise.	AI models showed promise but struggled with complex cases, indicating they should support, not replace, clinical judgment.
[25]	Age not reported	Hybrid DL model	Conditions studied included mild, moderate, and severe dental crowding based on arch length discrepancy	The hybrid deep learning model with unsharp masking performed best. Image enhancement improved efficiency but without significant differences. The model shows diagnostic potential but needs larger dataset validation.	The hybrid DL model outperformed DenseNet201 and EfficientNetV2 in accuracy and precision across all enhancement methods.
[26]	Age not reported	Deep learning	Crowded vs. noncrowded dentition (no Angle classification)	Deep learning, especially MobileNetV3 Small with CLAHE, accurately classified dental crowding (90.7% accuracy), outperforming junior orthodontists. Image enhancements like CLAHE boosted performance. Future work should use larger datasets and multiple image types.	MobileNetV3 Small with CLAHE consistently outperformed other models and junior orthodontists, showing strong potential for clinical decision support in identifying dental crowding from occlusal images.
[52]	18–60 years old	Deep learning based on Convolutional Neural Networks	Not specified	The program accurately detected nine facial measurements for automated orthognathic surgery analysis, though middle third and lower lip vermillion showed notable discrepancies.	The models were effective for automated facial analysis but limited by database size.
[42]	Age not reported	Deep learning-Fully Convolutional Neural Network	Class I, Class II (including half-cusp Class II), Super Class I, Class III, and unclassifiable	AI outperformed clinicians in malocclusion classification with high agreement but needs improvement in overjet/overbite measurements. Its multiphasic architecture is efficient but depends on strong initial training.	The AI outperformed clinicians in malocclusion classification but struggled with overjet and overbite measurements.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[97]	18–32 years old	Deep-learning	Skeletal Class III malocclusion	Models accurately and reproducibly analyzed condyles in Class III patients, reducing operator variability.	The models effectively segmented and analyzed remodeling, showing significant regional changes despite minimal overall alterations.
[98]	12–18 years old	Deep learning and convolutional neural networks	Classes I–III malocclusion	The AI provided faster, more consistent, and repeatable cephalometric analysis with reduced human error compared to manual tracing.	WeDoCeph results aligned more consistently with orthodontists, whereas WebCeph and CephX showed greater variability.
[47]	4–60 years old	Deep learning	Classes I–III malocclusion	Hyperparameter tuning and augmentation improved model accuracy, while explainable AI enhanced transparency and clinical trust.	The model's accuracy was 0.63–0.64 with hyperparameter tuning, improving 5–10% with augmentation; sensitivity ranged 0.59–0.65, specificity 0.80–0.81, and precision 0.61–0.62.
[27]	Age not reported	Deep learning, Convolutional Neural Network	Anterior crossbite	ResNet18 CNN accurately detects anterior crossbite in images and videos, supporting early malocclusion detection and remote screening.	ResNet18 CNN matched expert accuracy in classifying anterior crossbite from intraoral images and smartphone videos, offering high specificity and visualization-supported decisions.
[37]	14–55 years old	Deep learning model	Classes I–III malocclusion	The models boosted diagnostic accuracy, reduced manual effort, and improved orthodontic screening efficiency.	The SPMA network robustly enhanced orthodontic diagnosis by improving efficiency and accuracy in malocclusion classification.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[74]	12–25 years old	Vision-based machine learning	Spacing issues and misalignments but specific class of malocclusion not described	The model provides automated dental anomaly detection, reduces orthodontists' workload, and enhances diagnostic accuracy.	The AI system demonstrated high agreement with orthodontist evaluations, suggesting strong clinical applicability.
[78]	12–49 years old	Deep learning model, transformer models, and Minkowski convolutional neural networks	Focused on tooth and jaw relationships, but no specific class was assigned	The model allows 3D monitoring without extra radiation, faster and more efficient than manual analysis, improved patient safety, treatment accuracy, and potential for remote orthodontic monitoring.	The model developed a cross-temporal multimodal fusion system for orthodontic monitoring, achieving high segmentation accuracy and precision, significantly reducing clinical workload, and enabling future remote monitoring.
[35]	18 years and older	Deep learning model based on Transformer architecture (VSP transformer)	Classes I–III malocclusion	The VSP transformer accurately predicted jaw repositioning with errors of 1.34 mm—much lower than previous ~5 mm—incorporating soft tissue features for better performance and clinical interpretability, aiding surgical planning.	The VSP transformer achieved better generalization than other models. MAE: 1.34 mm; R^2 score: 0.5159 on the clinical test set. 94.89% of predictions within ±3 mm error margin.
[28]	18 years and older	Machine learning (non-neural)—Random forest and decision tree models	Class III malocclusion	The models efficiently identified key smile asymmetric parameters, helping prioritize treatment focus areas through machine learning.	The models analyzed smile changes post-surgical-orthodontic treatment for skeletal Class III malocclusion and identified factors affecting outcomes.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[21]	Age not reported	Deep learning model using a cascaded neural network	Classes I–III malocclusion	The surgery-first approach matched traditional methods in facial asymmetric correction, supported by reliable AI-assisted automated cephalometric analysis.	AI-assisted cephalometric analysis showed high agreement (interrater reliability 0.90) and effectively automated landmark tracing, reducing manual variability.
[99]	15 years and older	Web-based AI platform (WebCeph), deep learning-based automatic landmark detection	Classes I–III malocclusion	AI-based WebCeph offers reliable, efficient cephalometric analysis with clinically acceptable differences from manual measurements, though precision in complex landmarks needs improvement. It can streamline orthodontic workflows.	AI-based WebCeph provides clinically acceptable accuracy for most cephalometric measurements but requires improvement for condylion-related landmarks.
[92]	Age not reported	Deep learning-Convolutional neural network (CNN) for automated CBCT segmentation	Classes II–III malocclusion	AI improved CBCT analysis efficiency, reduced segmentation time, minimized interobserver variability, and enhanced surgical planning accuracy.	The AI-assisted workflow improved segmentation accuracy and enabled detailed condylar morphology analysis
[44]	18 years and older	Machine learning	General occlusal factors related to TMD; not classified by Angle Class	The Random Forest model using seven features accurately predicted TMD risk, supporting early clinical assessment and prevention.	The Random Forest model showed excellent TMD risk prediction in adults, with SHAP analysis enhancing interpretability for clinical use.
[100]	18 years and older	Machine learning-assisted	General malocclusion, not class-specific	The nomogram showed high predictive accuracy and clinical relevance, with an online tool enabling real-time TMD risk assessment for university students to guide early preventive interventions.	The constructed nomogram accurately predicts TMD risk in university students, supporting personalized orthodontic screening and early intervention.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[91]	10–45 years old	Deep learning	Maxillary Transverse Deficiency (MTD) focus; no Angle classification	The models demonstrated clinical applicability for faster and more reliable MTD diagnosis in orthodontics	The deep learning model accurately detected basal bone width landmarks with near-clinical precision, significantly reducing orthodontic diagnostic time.
[55]	16 years and older	Machine learning	Classes II–III malocclusion	Machine learning, especially combined models, predicts orthognathic surgery needs accurately; consensus diagnoses are easier to predict than final treatments, with Class III discrepancies predicted more reliably than Class II.	The combined ML approach successfully predicted orthognathic surgery necessity, performing better for Class III cases. Witts and ANB were the most influential cephalometric variables.
[45]	Age not reported	Deep learning facial landmark detection; rule-based heuristic classification	Class III malocclusion	AI-based skeletal Class III detection from profile images using heuristic rules is feasible, with a mobile-friendly model supporting early referral and screening.	Method 3 best balanced Class III detection and Class I/II misclassification, proving AI-driven malocclusion screening feasible using profile photos alone.
[79]	18–39 years old	Deep learning	Classes I–III malocclusion	IFOP may serve as a clinical predictor in orthodontic treatment planning for APMP-related cases.	IFOP inclination showed the strongest correlation with APMP indicators and was the most reliable predictor of mandibular positioning in deep learning 3D models.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[19]	16–22 years old	Rule-Based Decision Support System (non–machine learning)	Deepbite malocclusion (vertical discrepancy focus)	The rule-based AI DSS achieved 94.4% agreement with clinical treatments, maintaining 82% at the patient level, showing promise as a practical orthodontic decision support tool pending further validation.	The computerized DSS provided highly reliable and clinically acceptable deepbite treatment plans, closely matching actual orthodontic treatment outcomes
[41]	34–64 years old	Deep learning	Not specified (focused on segmentation for all dental anatomies, including cases with missing teeth and artifacts)	The model accurately segmented teeth, bone, and PDL spaces with robust performance on noisy CBCT data, enabling efficient tooth separation and direct use in finite element simulations.	The deep learning model accurately and reliably segmented teeth and bone structures, preserving critical periodontal ligament gaps suitable for realistic orthodontic finite element modeling
[56]	18–65 years old	Deep learning—Whisper AI speech detection, facial landmark tracking	General skeletal conditions	AI-based speech analysis is feasible and reliable for future clinical use in dental diagnostics.	AI systems effectively captured real-time mandibular movement during speech, revealing clear differences across occlusal parameters and age, showing strong clinical potential.
[101]	18–65 years old	Deep learning, AI-driven speech processing, and facial landmark tracking	Not stated	AI-based tracking reduced operator errors and enabled real-time, radiation-free speech articulation mapping, with potential in prosthodontic planning and early orthodontic assessment.	Reliable tool for tracking jaw and facial movement during speech. Robust performance despite not reporting traditional AI metrics.
[102]	Age not reported for patients studied	Tree-based models. Neural networks. Instance-based learning. Margin-based classifiers. Linear models	Angle Class I, II, III, Open Bite, Mandibular Deviation	ARIA demonstrated strong real-world performance, with extremely accurate, sensitive, and reliable diagnostics. There was stable output across 3000 cycles: low error rates in real-time conditions	ARIA demonstrated strong real-world performance, with extremely accurate, sensitive, and reliable diagnostics.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[75]	Age not reported for patients studied	Tree-based models. Neural networks, Instance-based learning. Margin-based classifiers, Linear models. Probabilistic models	The study classified the need for surgery, not to classify by traditional malocclusion types.	Machine learning reliably predicts orthodontic surgery needs, enabling faster and more consistent treatment planning.	Decision Tree and Gradient Boosting provided excellent results for classifying surgical needs, showing high accuracy, sensitivity, and robustness even with small sample size.
[43]	12–30 years old	Machine learning-Regression models. Support vector machine. Random forest	Skeletal Class II malocclusion	Machine learning models offer a new tool to assist orthodontists and patients in better planning treatment strategies.	The Random Forest model was the most effective and reliable method for predicting posttreatment facial esthetics in skeletal Class II camouflage extraction patients, providing clinically useful prediction accuracy.
[57]	14–50 years old	Deep learning-Convolutional neural networks, Dynamic graph convolutional networks	Various malocclusions, unspecified	DC-Net segments 3D intraoral scans faster and more accurately than prior methods. Dynamic local feature learning (EdgeConv) boosts digital dentistry, with plans to add boundary smoothing and extend to implants and prosthetics.	DC-Net enabled fast (24 s vs. 5–15 min), accurate, and reliable 3D tooth segmentation, significantly reducing manual labor and errors in orthodontic workflows.
[58]	Age not reported for patients studied	Deep learning-Convolutional Neural Network	Classes I–III malocclusion	Cut-out preprocessing plus CNN modeling is a promising direction for more efficient malocclusion diagnosis for Class I, Class II, and Class III	Cut-out preprocessing plus CNN modeling is a promising direction for more efficient malocclusion diagnosis.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[54]	6–12 years old	Deep learning models. nnU-Net. U-Net	General malocclusions (crowding, spacing, supernumerary teeth); no Angle Classifications	Model achieved expert-level classification and segmentation of CBCT scans, significantly improved workflow efficiency, strong robustness, and generalizability; future improvements needed for rare cases	AI significantly enhanced dental CBCT interpretation speed and accuracy, especially aiding junior dentists, with consistent performance across multiple clinical sites.
[29]	Age not reported for patients studied	Machine learning (non-neural)-Decision Tree, Random Forest, Support Vector Machine, Multilayer Perceptron	Malocclusions classified by Angle's and skeletal classification; crowding, overjet, overbite, incisor inclination, vertical growth pattern, facial profile considered	Machine learning enhances clinical decision support but cannot replace orthodontist expertise.	The Decision Tree reliably predicted extraction decisions; machine learning aids feature identification but should complement, not replace, clinical judgment.
[34]	10 years and older	Automatic segmentation model, unspecified deep learning architecture	General malocclusion patients were studied; no specific Angle Class (I, II, III) reported	A new 3D partitioning method using AI and intraoral scans was successfully developed to measure ERR.	AI-assisted methods reliably detected and quantified ERR in 3D with greater precision than traditional CBCT, offering valuable clinical orthodontic monitoring.
[32]	3–69 years old	Deep learning-Convolutional Neural Networks	Classes I–III malocclusion	Deep learning CNNs (especially Inception-ResNetV2) can accurately predict skeletal parameters from lateral profile photographs.	The CNN predicted skeletal discrepancies from non-radiographic images with reasonable accuracy, offering a radiation-free preliminary orthodontic screening tool.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[59]	Age not reported for patients studied	Deep learning-Convolutional Neural Networks, GoogLeNet and VGG-16 architectures	General dentofacial deformities; specific Angle Classifications (Class II, III) referenced but not separated in analysis	Deep learning models assist residents in diagnosing condylar OA from panoramic TMJ images, helping overcome inexperience in clinical settings.	Both deep learning models significantly outperformed inexperienced dental residents in diagnosing condylar OA on Con-Pa and Open-TMJ images ($p < 0.01$). Open-TMJ images provided the best diagnostic performance for deep learning models (AUC 0.89).
[66]	19–28 years old	Deep learning-Convolutional Neural Network	General dentofacial dysmorphosis including facial asymmetry, retrognathism, and prognathism	CNNs using facial photos effectively predict orthognathic surgery needs, aiding early skeletal screening but not replacing full clinical evaluations.	CNNs predicted orthognathic surgery needed from photos, highlighting lips, teeth, and chin as key discriminative features.
[85]	Age not reported for patients studied	Deep learning-DenseNet convolutional neural network	Skeletal Class I Class II and Class III	Deep learning CNNs can classify skeletal classes I, II, and III using cranio-spinal structures. DenseNet effectively extracts diagnostic features beyond the jawbone, enabling AI-driven classification without traditional landmarking.	CNNs classified skeletal classes using cranio-spinal features alone, with DenseNet effectively extracting key non-jawbone structures for accurate classification.
[86]	6–50 years old	Deep learning-Two-stage Convolutional Neural Networks	Various malocclusions; not separated by specific Angle Classifications	CephNet achieved highly accurate automatic cephalometric landmark detection with clinically acceptable errors (<1 mm), showing robustness across malocclusion types and imaging conditions, supporting clinical use in orthodontic planning and monitoring.	CephNet accurately localized cephalometric landmarks across different machines, malocclusions, and image qualities, improving orthodontic diagnostic efficiency.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[60]	Age not reported for patients studied	Deep learning-Convolutional Neural Network	Skeletal Class I, Class II, and Class III (based on ANB angle measurements)	ResNet-101 classifies skeletal malocclusions from cephalometric radiographs without manual landmarking, enhancing clinical efficiency; further external validation is needed.	The AI model accurately classified skeletal relationships without manual landmarks, enabling faster orthodontic diagnosis support.
[61]	10–40 years old	Deep learning-Convolutional Neural Network	Skeletal Class I, Class II, and Class III malocclusion	Deep learning (OCLU-NET) classifies skeletal occlusion from 3D dental scans more accurately than traditional models, supporting clinical orthodontic planning. Larger datasets and real-world testing are needed.	OCLU-NET outperformed traditional models in occlusion classification, showing strong potential to aid clinical orthodontic diagnosis despite preliminary status.
[38]	Age not reported for patients studied	Deep learning-Convolutional Neural Networks	Skeletal Class I, Class II, and Class III occlusions	Deep learning CNNs, especially Inception and DenseNet, accurately classify dental occlusion from 2D projections of 3D models, enabling automated, standardized diagnosis and faster treatment planning. Larger datasets and multicenter validation are required for clinical use.	Inception CNN balanced accuracy, precision, and recall best, reliably automating occlusion classification for clinical support.
[62]	7–25 years old	Distance-Weighted Discrimination classifiers	Skeletal Class III Malocclusion	Class III malocclusion subtypes were identified using statistical learning (DWD method); mandibular prognathic subtype had higher surgery and failure rates, while maxillary deficient subtypes responded better to nonsurgical treatment. The SPM3 model shows promise for early prognosis prediction.	The SPM3 model predicted Class III subtypes' surgical needs and treatment outcomes, enabling more personalized orthodontic care. treatment planning.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[33]	5–12 years old	Machine learning (non-neural) Logistic Regression, Random Forest, Gradient Boosting	Skeletal Class I, Class II Division 1, Class II Division 2, and Class III malocclusion	A mobile app using machine learning effectively pre-screens skeletal malocclusions from profile photos, aiding early Class III detection and timely consultation. Logistic Regression with bagging performed best. Broader validation across ethnicities and devices is recommended.	The ML-based mobile app effectively supports early malocclusion detection, especially Class III, but requires professional orthodontic confirmation after screening.
[103]	Age not reported for patients studied	Machine learning-Random Forest, Classification and Regression Trees, Conditional Inference Tree, Linear Discriminant Analysis, Support Vector Machine, K-Nearest Neighbor	Skeletal Class III Malocclusion	Machine learning models (Random Forest, CART, CTREE) accurately predict chin relapses after two-jaw surgery, with ramus inclination change as the key predictor, aiding personalized surgical planning and monitoring.	Random Forest, CART, and CTREE effectively predicted chin relapse post two-jaw surgery, with ramus inclination change—especially clockwise rotation over 3.7°—as the strongest relapse predictor.
[73]	6–8 years old	Clinical software (AI-assisted)	Classes I–III malocclusion	Digital cast analysis, manual or automatic, is clinically valid and faster than plaster methods. CS Model+ automates analysis, saving time but lacks accuracy in some categories. AI tools need further improvement for full clinical use.	Manual digital analysis aligns well with plaster models for most measurements, while automatic digital analysis is faster but less reliable for some categorical variables. midline alignment.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[48]	8–10 years old	Deep learning-Convolutional Neural Network	Skeletal Class I Malocclusion	The GP-GCNN model predicts short-term craniofacial hard-tissue growth with clinical precision, but chin-area soft-tissue predictions remain error-prone. AI growth models hold promise for early orthodontic planning but need broader validation, especially for Class II/III and diverse populations over longer follow-ups.	The AI-assisted GP-GCNN model accurately predicted most hard-tissue craniofacial growth but struggled with chin-area soft-tissue landmarks, needing further refinement.
[22]	0–40 years old	Machine learning-Random Forest, Logistic Regression, Support Vector Machine	Classes I–III malocclusion	Machine learning moderately predicts orthodontic extraction patterns from demographic and cephalometric data; larger datasets, clearer labels, and external validation are needed for improvement.	Random Forest best predicted U4 and U/L4 extractions, but machine learning struggled with less common extraction patterns.
[23]	16–67 years old	Two-Stage Mesh Deep Learning (TS-MDL) system	Class I malocclusion with anterior crowding or spacing	Deep learning automation (TS-MDL) accurately tracked 3D malalignment correction with Invisalign. While no single tooth movement predicted treatment time, combined factors and pretreatment PAR scores correlated strongly. AI-assisted imaging enhances orthodontic analysis efficiency and reliability.	Deep learning automation enabled precise tracking of malocclusion correction over time, minimizing manual effort and errors in Invisalign treatment monitoring.
[70]	Age not reported	Diffusion Probabilistic Models with MeshMAE and PointNet++ feature extraction	General malocclusion (crowding, spacing, overjet, overbite, etc.)	TADPM with diffusion probabilistic models enhances automated tooth arrangement by combining 3D mesh and point cloud features, reliably predicting final occlusion—even in severe cases—with clinical validation confirming orthodontic standards.	TADPM achieved state-of-the-art results and markedly improved clinical acceptability for automated tooth alignment planning over previous methods.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[88]	7–18 years old	Deep learning-Convolutional Neural Network	General malocclusion development; not limited to any one class	Dental arch dimensions vary by sex and region among Chinese adolescents, with peak growth at 13.7 years for males and 13.1 for females. Latitude correlates with arch width—northerners have wider arches. AI-assisted CNN measurement enables detailed orthodontic research, supporting region-specific treatment planning.	CNN-assisted segmentation reliably enabled large-scale 3D dental arch measurements, supporting valid growth curve analysis and regional comparisons.
[89]	20–70 years old	Artificial Intelligence–based statistical modeling system	General prosthetic cases	AI-designed zirconia crowns were clinically acceptable, reproducible, and matched standards for occlusal contact and marginal fit, reducing manual variation. Larger trials are needed for full validation.	AI-designed crowns showed high surface accuracy, good marginal fit, and clinically acceptable occlusion, matching digital designs with greater reproducibility.
[90]	18–35 years old	Rule-based geometry algorithm	Class III malocclusion	Automated design saved significant time, with AI and digital splints showing comparable surgical outcomes; vertical precision needs improvement for complex occlusions.	AI-generated splints matched manual digital splints in surgical simulation and live cases, reducing design time from minutes to about 10 s.
[104]	19–28 years old	Deep learning-Transfer Learning with Convolutional Neural Networks	Skeletal Class III	The machine learning model measured a 21% facial symmetry improvement post-orthognathic surgery. Xception CNN with constant-value augmentation showed top classification accuracy. The web system improved doctor-patient communication and surgical outcome consistency, with potential for real-time planning and broader facial assessments.	The model quantified a 21% average 3D facial symmetry improvement post-surgery, enhancing surgical evaluation and communication.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[46]	18 years old and up	Machine learning-Random Forest, AdaBoost, Multi-Layer Perceptron	Skeletal Class I, Class II, and Class III; Hypodivergent, Normodivergent, and Hyperdivergent	Machine learning, especially Random Forest, accurately discriminated skeletal discrepancies from 3D facial scans, with sagittal discrepancies identified better than vertical. PCA showed 87% of 3D facial variation is sagittal, highlighting soft tissue features as a promising non-invasive diagnostic tool.	Machine learning accurately classified skeletal discrepancies from 3D facial scans, with sagittal discrepancies easier to detect than vertical based on soft tissue morphology.
[63]	18 years old	Machine learning-Random Forest, Gradient Boosting, Decision Tree, SVM, K-Nearest Neighbors, Logistic Regression, Artificial Neural Network	Skeletal Class I, Class II, and Class III	Random Forest predicted skeletal Class I–III malocclusions with 74% accuracy from CBCT airway landmarks, achieving highest sensitivity for Class III (77%) and precision for Class II (80%). The method shows clinical promise but needs further validation.	Random Forest outperformed Gradient Boosting, Logistic Regression, Decision Tree, SVM, KNN, and ANN in predicting skeletal class using airway and cephalometric data.
[67]	7–44 years old	AI not specified	Classes II–III malocclusion	AI-assisted diagnosis could improve individualized orthodontic planning.	AI-assisted diagnosis could improve individualized orthodontic planning.
[53]	6–18 years old	Neural network	Classes I–III malocclusion	The CFOD AI system achieved 100% agreement with orthodontists on key malocclusion decisions, improved referrals (GPs 10×, pediatric dentists 2×), and proved efficient and reliable for managing orthodontic referrals in public healthcare.	There was 100% agreement with expert orthodontists for cases in the A-P, transverse, and vertical diagnostic planes.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[31]	4–14 years old	Deep learning, specifically 4 SOTA (state of the art) convolutional neural network (CNN) models	Classes I–III malocclusion	Deep learning models, especially DenseNet-121, accurately classified sagittal skeletal patterns in children, showing excellent performance for Class III malocclusions and potential for early orthodontic screening and treatment planning.	The DenseNet-121 AI program achieved over 90% accuracy and ~96.8% AUC with lateral cephalograms, showing strong potential as a reliable tool for diagnosing pediatric skeletal malocclusions.
[30]	Ages not reported for patients studied	Deep learning-CNN models (Conventional neural networks)	Frontal crossbite (KIG M4) and Lateral crossbite (KIG K4)	High crossbite detection accuracy on 2D photos, enabling remote diagnosis, virtual monitoring, and improved early referral by dentists and pediatricians.	CNN models (Xception, DenseNet, MobileNet) accurately detected crossbites on 2D intraoral images, with excellent binary classification performance and slightly lower accuracy for frontal vs. lateral cases, showing strong potential for AI-assisted orthodontic screening and triage.
[39]	Ages not reported for patients studied	AI-supported automated tracing technology designed for orthodontic applications.	Classes II–III malocclusion	AudaxCeph® AI reliably identifies cephalometric landmarks in severe Class II and III cases, though clinicians should verify less reliable points like Gonion and Porion. AI can streamline diagnostics, but oversight is essential.	AudaxCeph® showed clinically acceptable accuracy, with most landmark discrepancies within 2 mm of orthodontists' manual tracings, aside from a few exceptions.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[68]	5–49 years old	Machine learning (non-neural)-Support Vector Machines, K-Nearest Neighbors, Random Forest, Classification and Regression Trees, Linear Discriminant Analysis, and Generalized Linear Models	Classes I–III malocclusion	Machine learning, particularly a GLM using only SNA, SNB, and ML-NSL angles, reliably diagnosed skeletal Class I and III malocclusions in a German population, emphasizing mandibular sagittal and vertical positioning. The study recommended broader validation and integration of Class II diagnoses.	Certain models were able to achieve strong clinical accuracy
[64]	18 years old and up	Deep learning model based on a U-Net architecture	Class II malocclusion	A CNN-based AI model reliably predicted cephalometric changes after nonextraction Class II treatment, showing high accuracy for maxillary and incisor landmarks and moderate accuracy for mandibular changes. Heatmap visualizations aid treatment planning and patient communication. Future research should expand generalizability and include extraction cases.	AI-assisted outcome prediction holds strong orthodontic potential, needing broader validation and inclusion of extraction cases in future research.
[65]	21–26 years old	Back-propagation artificial neural network	Classes I–II malocclusion	A BP-ANN model reliably predicted pre-treatment dental and facial changes, with incisors easier to forecast than soft tissue. Key variables like lip position, Z angle, incisor inclinations, and facial convexity influenced aesthetics. Machine learning can aid personalized planning and extraction decisions.	The AI model achieved strong predictive accuracy.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[105]	4–22 years old	General machine learning techniques focused on feature selection and dimensionality reduction	Class III malocclusion	Machine learning combined with Boruta and LASSO effectively predicted Class III malocclusion progression, highlighting overlooked cephalometric variables like SN-PP and L1-MP angles as key prognostic tools for early risk stratification.	After ten-fold cross-validation, the model achieved 79.13% accuracy, with Boruta identifying L1-MP, PP-SN, and SNB angles as key predictors.
[[81]	13–27 years old	Linear regression and k-Nearest neighbor	These measurements are related to arch form and width, making the study relevant to malocclusion prevention but not tied to a specific malocclusion class like Class II or Class III.	k-Nearest neighbor effectively predicts dental arch measurements from incisor widths, enhancing diagnosis and personalized treatment to reduce anterior crowding. The study supports AI integration in orthodontics, urging broader validation for clinical use.	The k-Nearest neighbor algorithm outperformed linear regression, achieving ~99% accuracy with high correlation and minimal errors.
[106]	7–12 years old	Artificial neural network	Classes I–III malocclusion	Thumb sucking, especially its duration, strongly associates with anterior open bite development.	The AI tool achieved an ROC AUC of 0.889, 77.8% sensitivity, and 100% specificity, with no false positives and a 22.2% false negative rate.
[87]	14–18 years old	Convolutional neural networks	Classes I–III malocclusion	WebCeph™ and Cephio™ matched manual tracing for most measurements but require caution, especially for SNA, SN-PP, IMPA, and nasolabial angle. Automated AI tracing benefits diagnosis but needs refinement for critical landmarks to improve precision.	Automated cephalometric measurements by WebCeph™ and Cephio™ were clinically acceptable.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[107]	Ages not reported for patients studied	Deep learning	This study did not treat a specific class of malocclusion directly, but it was highly relevant to orthodontic and orthognathic surgical planning, including malocclusions typically addressed with cephalometric evaluations like Class II and Class III skeletal discrepancies	A lightweight deep learning model accurately and quickly detected diverse 3D cephalometric landmarks across varied datasets, showing clinical robustness and potential for integration into automated orthodontic and surgical planning to enhance efficiency and standardization.	The AI system had mean localization errors of ~1.96 mm (Finnish) and ~1.99 mm (Thai), with 61.7–64.3% of landmarks within 2 mm clinically acceptable limits; cephalometric characteristic accuracy was even higher.
[108]	20–26 years old	Software-assisted landmark autodigitization system	Class III malocclusion	Cleft patients showed less predictable soft tissue movement, posteriorly positioned landmarks, and wider nasal structures compared to controls. Though bimaxillary surgery improved esthetics in both groups, cleft patients' unique responses highlight the need for adjunctive procedures. AI-assisted 3D analysis aids in detecting these nuanced outcomes.	AI-assisted ON3D software reliability confirmed by consistent landmark identification in repeated digitization, though conventional metrics like accuracy and sensitivity were not reported.
[49]	9–12 years old	Deep learning-based 3D convolutional neural network	Class II malocclusion	Both Twin Block and Functional Regulator II appliances effectively stimulated mandibular and condylar growth in skeletal Class II cases, with no significant differences in growth, volume, or positional changes, indicating comparable effectiveness.	In terms of performance, the 3D UX-Net segmentation achieved good accuracy. The AI- based segmentation also allowed for rapid model generation.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[109]	19–29 years old	Convolutional neural networks	Classes II–III malocclusion	A deep learning model accurately predicted the need for orthognathic surgery from cephalometric images, showing strong potential as a screening tool in dental and surgical practice.	The AI tool achieved 95.4% accuracy, 84.4% sensitivity, and 99.3% specificity, demonstrating strong performance.
[110]	Ages not reported for patients studied	A modified convolutional neural network architecture based on SqueezeNet.	Classes I–III malocclusion	The DB4 Dental Classifier and Smart Search Engine outperformed Google in dental image recognition and classification, showing high accuracy and strong clinical and research potential.	The AI program achieved 93% categorical accuracy using the DB4 Smart Search Engine, with top-k categorical accuracy of 100% in training and validation, indicating excellent performance.
[111]	Ages not reported for patients studied	A fully convolutional deep neural network called the "You Only Look Once" (YOLO) model	Classes I–III malocclusion	The AI engine accurately detected and localized orthodontic issues from intraoral images, showing promise for orthodontic screening and expanding access through automated assessments.	The AI program achieved 99.99% accuracy, 99.79% precision, 100% recall, and an F1 score of 1.00, demonstrating excellent performance.
[20]	7–12 years old	Artificial neural network	Classes I–III malocclusion	Thumb sucking and its duration were strongly linked to anterior open bite in children, while other factors showed no significance in the ANN model.	The AI tool achieved an AUC of 0.889, 77.8% sensitivity, 100% specificity, no false positives, and a 22.2% false negative rate.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[77]	16–29 years old	Logistic Regression, Support Vector Machine (SVM), Multilayer Perceptron (MLP), k-Nearest Neighbor (kNN), Random Forest, Convolutional Neural Network (CNN), and Extreme Gradient Boosting (XGBoost).	Class III malocclusion	Key cephalometric measures—Wits appraisal, overjet, and Mx/Md ratio—were most influential in distinguishing surgical from non-surgical needs in adult Class III cases, aiding treatment planning.	The XGBoost model showed strong predictive reliability for both treatment groups.
[76]	12–68 years old	Decision tree algorithms	Classes I–III malocclusion	AI decision processes improved patient compliance and behavior, while dental monitoring tools proved valuable for remote evaluation.	The AI tool improved patient interaction and discipline but showed no significant aligner tracking improvement for men.
[72]	Ages not reported for patients studied	Symbolic AI and machine learning	Classes I–II malocclusion	SmileMate AI showed only slight to moderate agreement with clinical assessments, making it unreliable for diagnosing malocclusions without clinician supervision.	The performance of the AI program in terms of outcomes showed an overall sensitivity of 72% and specificity of 54%.
[40]	Ages not reported for patients studied	Convolutional neural networks (CNNs) and Transformers	Classes I–III malocclusion	The study highlighted AI's potential to revolutionize orthodontics by streamlining design and improving outcomes, while noting the need for further research in AI-assisted tooth alignment.	The AI tool showed promise in enhancing the efficiency and accuracy of orthodontic treatment planning, paving the way for further advancements in AI-assisted orthodontics.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[112]	Ages not reported for patients studied	XGBoost, AdaBoost, and ExtraTrees, as well as linear regression models.	Classes I–III malocclusion	Nine cephalometric features—covering tooth position, jaw alignment, and soft tissue morphology—significantly influenced expert evaluations. Machine learning showed potential to enhance accuracy and efficiency in orthodontic treatment assessment.	The XGBoost model achieved an MAE of 0.267, RMSE of 0.341, and Pearson's r of 0.683 with nine features, indicating high predictive accuracy.
[71]	14–45 years old	Not specified	Classes I–III malocclusion	Automated digital setup systems improved efficiency and showed clinical potential but still need refinement to match manual precision.	AI tool performance was evaluated against manual setups using linear and angular movement errors and the Peer Assessment Rating index.
[69]	11–16 years old	Least squares regression, ridge regression, lasso regression, elastic net regression, XGBoost, random forest, and a neural network	Class I malocclusion	Early mandibular length, maxillary length, and lower face height best predicted post-pubertal mandibular length; Y-axis growth was predicted by prior Y-axis values, lower face height, and mandibular plane angle. Machine learning effectively predicted craniofacial growth, aiding orthodontic planning.	AI models predicted mandibular length with 95.8–97.6% accuracy and Y-axis growth with 96.6–98.3% accuracy.
[113]	Ages not reported for the patients studied	Deep learning	Not listed	AI can predict the maxillomandibular relationship from digitized teeth scans with reasonable accuracy, but clinical use is limited by errors above acceptable thresholds and reliance on stone cast–based training.	The AI showed discrepancies under 1.5° and 1.3 mm—feasible, but not yet clinically acceptable.

Author (Year)	Study Population	AI Type	Condition	Clinical Application	Outcomes and Notes
[36]	12–65 years old	Convolutional neural networks	Classes I–III malocclusion	The study showed that lateral photographs can classify skeletal malocclusions without radiography, with AI outperforming orthodontists, especially in severe cases, and offering potential for early detection and self-health management in family medicine.	Model5 achieved 84.5% accuracy, 77.32% sensitivity, and 88.44% specificity, with all metrics for Class III malocclusion exceeding 90%.
[24]	8–14 years old	Convolutional neural networks	Class III malocclusion	The deep learning model accurately predicted mandibular growth trends, outperforming junior orthodontists, by focusing on key cephalometric regions—the chin, mandible edge, incisor area—and unexpectedly, the airway, revealing new research opportunities.	The AI tool achieved 85% accuracy, 0.95 sensitivity, 0.75 specificity, and a 0.9775 AUC, outperforming junior orthodontists.
[114]	4–14 years old	Gaussian Process Regression, Radial Basis Function Support Vector Machine, Quadratic Discriminant Analysis, Linear SVM, and others. Recursive Feature Elimination	Class III malocclusion	Machine learning, especially Gaussian Process Regression, can effectively classify pediatric Class III malocclusions using key cephalometric features—SN-GoMe, U1-NA, Overjet, and ANB—highlighting AI's potential to improve early orthodontic diagnosis and treatment planning.	The GPR model achieved the best performance, with an AUC value of 0.879. Specific accuracies for dental, skeletal, and functional classifications varied, with dental Class III malocclusions achieving the highest accuracy at 87.50%.

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